

PROFESSIONAL SUMMARY

Recent B.Tech Computer Science graduate with hands-on experience in computer vision and deep learning (transfer learning, CNN-based classification), alongside full-stack web development, through academic projects. Proficient in Python, React.js, Flask, TensorFlow, and PostgreSQL, with a strong foundation in data structures, databases, and software engineering.

EDUCATION

Graphic Era Hill University, Dehradun
B.Tech in Computer Science and Engineering

Aug 2022 – May 2026
CGPA: 7.22

TECHNICAL SKILLS

Languages: C, C++, Java, Python, SQL

Web Development: HTML, CSS, JavaScript, React.js, Flask

ML/DL Libraries: TensorFlow, Keras, OpenCV, NumPy, MTCNN

ML/DL Concepts: Computer Vision, CNN Architectures, Transfer Learning

Databases: SQLite, PostgreSQL

Tools: Git, GitHub, VS Code, JupyterLab, Google Colab

PROJECTS

DeepFake Detection System | Python, EfficientNet-B4, MTCNN, TensorFlow, OpenCV

[GitHub](#)

- Fine-tuned a pretrained EfficientNet-B4 backbone via transfer learning for real/fake classification.
- Applied MTCNN-based face detection and cropping to isolate facial regions, improving classifier signal.
- Built a Flask REST API for image and video uploads with real-time confidence scores.
- Optimized pipeline to analyze images in under 3 seconds and videos in under 30 seconds.
- Analyzed videos by sampling 30 evenly-spaced frames and aggregating per-frame predictions into a verdict.

Ultimate Tic-Tac-Toe | React.js, Vite, Supabase, Web Audio API

[GitHub](#) | [Live Demo](#)

- Designed an algorithm to procedurally generate win conditions for arbitrary grid sizes (3x3–5x5).
- Implemented a Minimax AI with alpha-beta pruning across four selectable difficulty levels.
- Built real-time multiplayer using Supabase Broadcast channels for live move sync and game invites.
- Implemented an MMR-based ranked matchmaking system with grid-size-scaled scoring.
- Built a synthetic audio engine with the Web Audio API, using oscillators for sounds and win/draw jingles.

Air Quality Predictor | React.js, FastAPI, Random Forest, SQLite

[GitHub](#) | [Live Demo](#)

- Built a full-stack ML app using Random Forest to forecast PM2.5 levels with an R^2 of 0.89.
- Integrated Open-Meteo APIs with FastAPI and SQLite for real-time air quality data.
- Developed a React dashboard with Leaflet maps and Chart.js visualizations.
- Implemented an SQLite caching layer to store API responses and log search history, cutting redundant calls.

CERTIFICATIONS

- Completed the Cybersecurity Job Simulation Analyst certification.
- Earned the Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate certification.
- Completed Introduction to Generative AI by Google Cloud & Simplilearn SkillUp.

COURSEWORK SKILLS

- Data Structures and Algorithms (DSA)
- Machine Learning (ML)
- Database Management Systems (DBMS)
- Object-Oriented Programming (OOP)
- Computer Networks (CN)
- Software Development Life Cycle (SDLC)
- Operating Systems (OS)